Code No: 126AP **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, April - 2018
DISTRIBUTED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

Used paper 27-04-2018

PART - A

(25 Marks)

1.a)	Define and explain about the distributed systems.	[2]
b)	What is mean by resource sharing? Explain.	[3]
c)	Present a note on external synchronization.	[2]
d)	Write about election algorithm.	[3]
e)	Differentiate unicast and multicast communication.	[2]
f)	Write a short note on group communication.	[3]
g)	Discuss about distributed shared memory.	[2]
h)	What are the requirements of the distributed file systems?	[3]
i)	What is deadlock? Explain.	[2]
j)	Write about two phase locking.	[3]

PART - B

(50 Marks)

2.a) Explain about architectural elements.
b) Write a short notes on characteristics of distributed systems. [5+5]

OR

3. Explain in brief about system models of distributed systems. [10]

4.a) Discuss about distributed mutual exclusion.
b) Discuss about consensus and related problems. [5+5]

OR

5.a) Explain about clocks, events and process states.
b) Discuss about global states. [5+5]

6.a) Discuss about the API for the Internet protocols.
b) Explain about IPC in UNIX. [5+5]

OR

7.a) Discuss about communication between distributed objects.
b) What is a Remote Procedure Call(RPC)? Explain. [5+5]

8.a) Explain about Andrew file system.
b) Explain about design and implementation issues of distributed shared memory. [5+5]

OR

9. Explain the following.
a) Directory services.
b) Release consistency in distributed shared memory. [5+5]

10.a) What is mean by atomic commit protocols? Explain.
b) Discuss about timestamp ordering. [5+5]

OR

11.a) Discuss about concurrency control in distributed transactions.
b) Explain about flat and nested distributed transactions. [5+5]

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